



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION, SEMESTER II – DECEMBER 2024
STA1023/MAT1063 – INTRODUCTION TO PROBABILITY THEORY

Time Allowed: **TWO Hours**

Answer **All** Questions.

1

- a). Prove the following identity. [20 marks]

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

- b). A student has 3 mathematics books, 4 history books, 3 chemistry books and 2 biology books.

- i) If the student selects four books randomly, what is the probability that all four books are from different subjects? [15 marks]
- ii) If the student selects three books randomly, what is the probability that there are at least two mathematics books? [15 marks]
- iii) In how many ways can these books be arranged on a shelf so that all books of the same subjects are together? [20 marks]

- c). Suppose a fair n -sided die is rolled n independent times. A match occurs if side i is observed on the i^{th} trial, $i=1,2,3,\dots,n$. Show that the probability of at least one match is

$$1 - \left[\frac{n-1}{n} \right]^n \quad [30 \text{ marks}]$$

2. A continuous random variable X has probability density function:

$$f_X(x) \begin{cases} ax & ; \quad 0 \leq x \leq 8 \\ 8a & ; \quad 8 < x \leq 9 \\ 0 & ; \quad \text{otherwise} \end{cases}$$

- a). Find the value of a such that $f(x)$ is a legitimate pdf. [15 marks]
- b). Derive the Cumulative Density Function (CDF) of X . [20 marks]
- c). Calculate $P(X > 7)$. [20 marks]
- d). Derive an expression for the r^{th} moment about the origin for X . [20 marks]
- e). Calculate $E(X)$ and $Var(X)$ using the expression in (d). [25 marks]

[P.T.O.]

3.

a). Let X be a discrete random variable where the moment generating function, $M_x(t)$ exists.

i) For constants a and b , show that in usual notation, [20 marks]

$$M_{\frac{(x+a)}{b}}(t) = e^{\left(\frac{a}{b}\right)t} M_x(t/b)$$

ii) If the random variable X has the Binomial Distribution with n trials and p probability of success, show that the moment generating function (MGF) of X is

$$M_x(t) = \frac{(pe^t)^n}{[1-(1-p)e^t]^n} \quad [20 \text{ marks}]$$

iii) Use the MGF to find the mean value of X . [20 marks]

b). Customers arrive in a certain shop according to an approximate Poisson process at a mean rate of 25 per hour.

i) What is the probability that the shopkeeper will have to wait exactly 5 minutes for the arrival of the first customer? [10 marks]

ii) Find a lower bound for the probability of waiting time within two standard deviations of mean. State the inequality you used to obtain the answer. [15 marks]

iii) Use Normal Approximation to calculate the exact probability value indicated in part (ii). [15 marks]

4.

a). Suppose 10 cards are drawn one at a time with replacement from a standard deck of playing cards. What is the probability that a Diamond card will be picked

i) exactly 3 times? [15 marks]

ii) for the first time in the 4th roll? [15 marks]

iii) for the 4th time in the 7th roll? [20 marks]

b). The speeds of buses of a certain brand are normally distributed with a mean of 90 km/hr and a standard deviation of 10 km/hr.

i) What percentage of buses will ride within speed 87km/hr and 102 km/hr? [15 marks]

ii) Calculate the 90th percentile for speed. [15 marks]

iii) What percentage of buses will fall within one standard deviation of the mean speed.

[20 marks]

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Distribution	Density Function
Binomial (n, θ)	$P(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}; x = 0, 1, 2, \dots, n$
Negative Binomial (k, θ)	$P(X = x) = \binom{x-1}{k-1} \theta^k (1 - \theta)^{x-k}; x = k, k + 1, k + 2, \dots$
Geometric (θ)	$P(X = x) = \theta (1 - \theta)^{x-1}; x = 1, 2, \dots$
Poisson (λ)	$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}; x = 0, 1, 2, \dots$ where $E(X) = \lambda, Var(X) = \lambda$